

CF06 CMAN

Kay Hynes

a) Ideal read

$$I_{\text{col0,ideal}} = \frac{V_{\text{row0}} - V_{\text{col0}}}{R[0][0]} = \frac{1 - 0}{1 \text{ k}\Omega} = 1 \text{ mA}$$

b) Floating-node solution

Applying KCL at the floating row node gives:

$$\begin{aligned}\frac{V_{\text{row1}} - 0}{2 \text{ k}\Omega} + \frac{V_{\text{row1}} - V_{\text{col1}}}{1 \text{ k}\Omega} &= 0 \\ V_{\text{row1}} + 2V_{\text{row1}} - 2V_{\text{col1}} &= 0 \\ 3V_{\text{row1}} - 2V_{\text{col1}} &= 0\end{aligned}$$

Applying KCL at the floating column node gives:

$$\begin{aligned}\frac{V_{\text{col1}} - 1}{2 \text{ k}\Omega} + \frac{V_{\text{col1}} - V_{\text{row1}}}{1 \text{ k}\Omega} &= 0 \\ V_{\text{col1}} - 1 + 2V_{\text{col1}} - 2V_{\text{row1}} &= 0 \\ 3V_{\text{col1}} - 2V_{\text{row1}} &= 1\end{aligned}$$

Solving the two equations together:

$$\begin{aligned}\begin{bmatrix} 3 & -2 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} V_{\text{row1}} \\ V_{\text{col1}} \end{bmatrix} &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ \begin{bmatrix} V_{\text{row1}} \\ V_{\text{col1}} \end{bmatrix} &= \frac{1}{5} \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ V_{\text{row1}} &= \frac{2}{5} = 0.4 \text{ V} \\ V_{\text{col1}} &= \frac{3}{5} = 0.6 \text{ V}\end{aligned}$$

c) Sneak-path current

Selected-cell current:

$$I_{00} = \frac{1 - 0}{1 \text{ k}\Omega} = 1 \text{ mA}$$

Floating row injects current into column 0 through $R[1][0]$:

$$I_{10} = \frac{V_{\text{row}1} - V_{\text{col}0}}{R[1][0]} = \frac{0.4 - 0}{2 \text{ k}\Omega} = 0.2 \text{ mA}$$

So, actual sensed current:

$$I_{\text{col}0, \text{actual}} = I_{00} + I_{10} = 1 \text{ mA} + 0.2 \text{ mA} = 1.2 \text{ mA}$$

d) MVM corruption

In an ideal MVM read, each column current should equal the sum of the driven row voltages weighted by the programmed cell conductances. Sneak paths corrupt that sum by adding unintended current through unselected cells. In larger arrays this can add significant error, requiring mitigation such as biasing schemes, isolation devices, etc.